Cisco 8000 Series Routers Datasheet

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OVERVIEW

The Cisco 8000 Series combines Cisco's revolutionary Cisco Silicon One, IOS XR software, and a set of clean-sheet chassis to deliver a breakthrough in high-performance routers. The 8000 Series comprises a full range of feature-rich, highly scalable, deep-buffered, 400G-optimized routers ranging from 10.8 Tbps in a 1 RU footprint to an industry-leading, rack-mountable modular system capable of ~260 Tbps of full duplex, line rate forwarding in its initial release.

The Cisco 8000 Series includes two distinct router architectures that both utilize the Cisco Silicon One ASIC. The 8800 Series provides the highest bandwidth via modular chassis with a redundant control plane and switch fabric. The 8800 Series includes the Cisco 8808, Cisco 8812, and Cisco 8818. These chassis deliver up to 14.4 Tbps per line card via 100 and 400 Gigabit Ethernet (GbE) ports.

The Cisco 8200 Series utilizes Cisco's new Router-on-Chip (RoC) model to deliver full routing functionality with a single ASIC per router. The RoC architecture is distinguished from System-on-Chip (SoC) switches by supporting large forwarding tables, deep buffers, more flexible packet operations, and enhanced programmability. Both the 8201 and 8202 provide 10.8 Tbps of network bandwidth with dramatically lower power than contemporary 10 Tbps systems.

CISCO 8000 SERIES HARDWARE

The 8000 Series includes five routers to address a broad range of bandwidth needs and facility requirements.

Table 1. Cisco 8000 Series chassis options

Chassis	Bandwidth	Ports	Height
Cisco 8201	10.8 Tbps	24 QSFP56-DD 400 GbE 12 QSFP28 100 GbE	1 RU
Cisco 8202	10.8 Tbps	12 QSFP56-DD 400 GbE 60 QSFP28 100 GbE	2 RU
Cisco 8808	115.2 Tbps	48 QSFP28 100 GbE with MACSec	8 slots / 16 RU

		36 QSFP56-DD 400 GbE	
Cisco 8812	172.8 Tbps	48 QSFP28 100 GbE with MACSec 36 QSFP56-DD 400 GbE	12 slots / 21 RU
Cisco 8818	259.2 Tbps	48 QSFP28 100 GbE with MACSec 36 QSFP56-DD 400 GbE	18 slots / 33 RU

SILICON INNOVATION WITH THE CISCO SILICON ONE ASIC

At over twice the network capacity of all other high-scale ASICs, Cisco Silicon One is the first routing silicon to break through the 10 Tbps benchmark for network bandwidth. This is accomplished without sacrificing route capacity, packet-per-second forwarding performance, or feature flexibility.

To achieve this unprecedented performance, one of the most challenging factors limiting capacity was addressed. High-scale routing and deep buffering typically require off-chip memories. Connecting an ASIC to these memories reduces the bandwidth available to the data path. This is one of the key reasons that routing chips traditionally have less bandwidth than SoC designs. The Cisco Silicon One architecture achieves high-performance and full routing capabilities without external memories. This is enabled by the clean-sheet internal architecture that includes an on-chip High Bandwidth Memory (HBM). Originally developed for high-end graphics, HBM provides a significant increase in memory performance while lowering power consumption. It is located on the chip package and connects to the forwarding logic via an ultra-fast silicon interface. In Cisco Silicon One ASIC, the HBM is used for both deep buffering and to extend the forwarding tables, thus eliminating the need for external memories.

The Cisco Silicon One architecture supports multiple modes of operation. It can function as an RoC, a line card network processor, and a switch fabric element. This flexibility enables consistent software in multiple roles and rapid silicon evolution.

SYSTEM DESIGN INNOVATION

Supporting the 8000 Series capabilities demanded a wide range of new approaches to platform design. By leveraging over 25 years of high-performance system design, Cisco has delivered unprecedented

capacity without compromising forwarding performance or requiring oversubscription. This required new power supplies, a redesigned cooling architecture, and future-proof connectors.

The Cisco 8200 Series chassis requires extensive innovation to support high densities. Consolidating multiple components into a single ASIC requires significant chip power. In addition, 400 GbE optics require up to six times the power of 100 GbE QSFP28 modules. These challenges are addressed via advanced system design, including state-of-the-art fans and heat sinks, and QSFP-DD modular optics.

The 8800 chassis utilize a state-of-the-art orthogonal direct design with advanced cooling, high power capacity, new power supplies, and future-proof connectors. The chassis and all data path components for the 8800 Series benefit from a clean sheet design that allows the systems to take full advantage of the latest technologies and Cisco's design expertise. This design connects all forwarding path components directly without a backplane or midplane. In the 8800 Series, the line cards are oriented horizontally, and the eight fabric cards are oriented vertically. Every major component of the 8800 Series was developed with a clean-sheet approach – representing Cisco's unprecedented investment and commitment to a long lifecycle for the 8000 Series.

The 8800 chassis deliver significant improvements over previous orthogonal chassis, including:

- State-of-the-art redundant fans
- Network Equipment Building System (NEBS)-compliant air filters with doors for simplified line card access
- Future-proof power capacity with power-saving internal distribution
- New power supplies for power feed redundancy with reduced provisioning
- Cable management for up to 864 fibers

APPEARANCE

Figure 1. Cisco 8201 Router

Figure 2. Cisco 8202 Router

Figure 3. Cisco 8812 Router

Figure 4. 36-port QSFP56-DD 400 GbE line card

Figure 5. 48-port QSFP28 100 GbE line card

Figure 6. Cisco 8812 fabric card

SECURITY

Security is a major concern for all Cisco customers. Attacks on networking equipment can have disastrous results. Network operators need assurance that their equipment is secure and running authorized Cisco software. Cisco 8000 Series routers support hardware root of trust based on the Trusted Computing Group (TCG) and IEEE 802.1 AR standards. This approach is far more reliable than a software-based security approach. All Cisco 8200 routers are FIPS 140-2 Level 2 compliant and support advance security features to ensure platform and OS integrity.

- Cisco secures the supply chain of every system at manufacturing time. A technology called "Chip Protection" allows customers to be assured that the hardware they receive from Cisco has no counterfeit components. This is accomplished with the use of unique identifiers that are stored inside Trusted Anchor Module (TAM) device as a way to identify and track componenets through the entire lifecycle of Cisco 8000. The checks cover all major components, including network processors, CPUs, and Field-Programmable Gate Arrays (FPGAs).

- Every image a customer downloads from Cisco site is cryptographically signed using Cisco private keys. Each platform has a TAM (based on the TCG standard) that uses built-in cryptographic functions to validate the image signature. Once the signature is validated, the software is confirmed authentic and is ready for install.

- During normal operation, the JTAG (Joint Test Action Group – a method of chip testing and verification) ports on chips are monitored. JTAG is one of the most common attack surfaces and

therefore must be secured. Cisco uses a technology called "Secure JTAG" to monitor the port. If any illegal activity is detected, it is flagged and the system CPU is held in reset mode.

- Secure Boot root of trust is anchored in the TAM. It establishes an authentication chain where each software module authenticates the next module in the boot process.

CISCO IOS XR SOFTWARE

Cisco IOS XR7 is a unified network OS spanning access, aggregation, edge, and core. The networking protocol stack within XR7 can be cut down by two-thirds when the IP transport architecture is simplified. Improvements to XR7 internal architecture have reduced the memory footprint by 35%. By reducing code size and the resources required, XR7 can be installed onto even the most constrained hardware designs with full security features without impact to boot times.

Modernizing XR7 with install procedures using standard Linux software package managers has also improved operations. Instead of "one-size-fits-all," XR7 provides modularity so customers only load what they will use. Service providers can easily access new software packages from trusted Cisco Red Hat Package Manager (RPM) repositories. Alternatively, they can build their own repository of both Cisco and custom software packages, which can be fetched for final system configuration without spending time trying to sort out software dependencies. All the required Cisco software packages, any home-grown/third-party software packages, as well as the router configuration can be pulled into a single Cisco software image known as a "Golden ISO." Customized images can now be installed consistently and with confidence across devices in the network.

Cisco IOS XR7 brings an unmatched level of openness for programmability and customization.

IOS XR 7 supports open, model-driven APIs at all layers of the software stack. At the management layer, XR supports a comprehensive list of both native and industry-driven OpenConfig models with multiple encoding (XML and JSON) and transport (gRPC, Netconf) options. The APIs at the management layer allow operators to apply configuration to the device or retrieve the state of the system. The APIs also address advanced traffic engineering use cases, allowing applications to control the route followed by traffic within the network. These APIs can be used independently or combined with other ecosystem abstraction layers such as SONIC or P4Runtime.

IOS XR 7 also supports the OFA (Open Forwarding Abstraction) API, which provides a logical representation of all the forwarding and telemetry capabilities of the underlying hardware. In addition, IOS XR 7 provides a flexible consumption model allowing third-party application software to run on the device alongside IOS XR to enable customization options for the customer network. With application hosting capabilities, operators can host their own controller agent or custom protocol; use

various hosting apps for telemetry collection, traffic engineering, and configuration management; or manage the box like a Linux machine using third-party software such as Chef, Puppet, or Ansible.

Cisco IOS XR 7 is the industry's most trusted network operating system.

XR7 is the most advanced network operating system for improving the security posture of the router. The Cisco Secure Boot subsystem ensures that the device boot image is genuine and untampered. With advanced signing technology, XR7 can establish software integrity enforcement and measurement. To further enhance the trusted defense posture, multiple runtime defenses within XR7 guard against malicious actors and make exploitation of bugs more difficult. Even if booted securely, a router may run for months or years without rebooting, which could leave vulnerabilities at runtime undetected for a long time. XR7 leverages Integrated Measurement Architecture (IMA) to significantly enhance security by verifying the integrity of running software. In the IMA appraisal mode, signature validations prevent unauthorized images to launch. In the IMA measurement mode, the hashes of all images are logged in a secure location used for verification. Records of run time processes can be sent for analysis, so the operator knows that system software, updates, or patches are running as intended.

BASEBOARD MANAGEMENT CONTROLLER

The Baseboard Management Controller (BMC) provides specialized controllers that monitor the state of a computer or network equipment. It enables monitoring and management of various aspects of the system, such as health (for example, temperature and voltages) and log events for failure analysis. It also provides a range of other remote management capabilities, such as environmental control and firmware upgrades, even when the main CPU is rebooting or unavailable.

PHYSICAL SPECIFICATIONS

Table 2. Physical Specifications

Series or model	Physical characteristics
Cisco 8000 Series	Operating temperature: 32 to 104° F (0 to 40° C) Nonoperating temperature: -40 to 158° F (-40 to 70° C) Humidity: 5% to 95% (noncondensing) Altitude: 0 to 9842 ft. (0 to 3000 m)

Cisco 8200 Series	Intel Broadwell 4-core 2.4 GHz CPU with 32 GB of DRAM. RS-232 console, 10 GbE Control Plane expansion, 1 GbE Management & BMC port, 1x USB2.0, GBP (ToD, 10MHz, 1PPS), 1588, and BITs (sync). Bidirectional air flow
Cisco 8201	(H) 1.73 x (W) 17.3 x (D) 20.1 in. (4.40 x 43.9 x 51.1 cm) – 1 RU 24 lbs (10.9 kg) Typical system power at 10.8 Tbps: 415 W 2 power supplies
Cisco 8202	(H) 3.45 x (W) 17.3 x (D) 20.1 in. (8.77 x 43.9 x 51.1 cm) – 2 RU 42 lbs (19 kg) Typical system power at 10.8 Tbps: 750 W 2 power supplies Optional air filter
Cisco 8800 Series	2 route processors Intel Broadwell 4-core 2.4 GHz CPU with 32 GB of DRAM. RS-232 console, 10 GbE Control Plane expansion SFP+, 1 GbE Management & BMC port, 2x USB2.0, GBP (ToD, 10MHz, 1PPS), 1588, and BITs (sync). Cable management, doors, and air filters 6.3 KW power supply for AC and high-voltage DC (180 to 305V AC, 192 to 200V DC) 4.4 KW power supply for 48V 60A DC (-40 to -75V DC) 4.8 KW power supply for 48V 100A DC (-40 to -75V DC) Horizontal line cards with 8 vertical fabric cards and 4 fan trays Rack mountable in standard 19 in. (48.3 cm) rack

Cisco 8800 Series	2 route processors Intel Broadwell 4-core 2.4 GHz CPU with 32 GB of DRAM. RS-232 console, 10 GbE Control Plane expansion SFP+, 1 GbE Management & BMC port, 2x USB2.0, GBP (ToD, 10MHz, 1PPS), 1588, and BITs (sync). Cable management, doors, and air filters 6.3 KW power supply for AC and high-voltage DC (180 to 305V AC, 192 to 200V DC) 4.4 KW power supply for 48V 60A DC (-40 to -75V DC) 4.8 KW power supply for 48V 100A DC (-40 to -75V DC) Horizontal line cards with 8 vertical fabric cards and 4 fan trays Rack mountable in standard 19 in. (48.3 cm) rack
Cisco 8808	(H) 28 x (W) 17.45 x (D) 33.73 in. (71.12 x 44.32 x 85.7 cm) – 16 RU – 8 line cards Depth with cable management, filter, and doors: 41.18 in. (104.6 cm) Weight: Unloaded, 162 lbs (73 kg); fully loaded, 643 lbs (292 kg) Typical system power at 115.2 Tbps: 13 KW 9 high-voltage power supplies or 12 48V DC power supplies
Cisco 8812	(H) 36.75 x (W) 17.45 x (D) 35.43 in. (93.345 x 44.23 x 90 cm) – 21 RU – 12 line cards Depth with cable management, filter, and doors: 41.55 in. (105.5 cm) Weight: Unloaded, 212 lbs (96 kg); fully loaded, 868 lbs (393 kg) Typical system power at 172.8 Tbps: 20 KW 9 high-voltage power supplies or 12 48V DC power supplies
Cisco 8818	(H) 57.75 x (W) 17.45 x (D) 35.43 in. (146.7 x 44.23 x 90 cm) – 33 RU – 18 line cards

	Depth with cable management, filter, and doors: 41.55 in. (105.5 cm) Weight: Unloaded, 283 lbs (128 kg); fully loaded, 1323 lbs (600 kg) 18 high-voltage power supplies or 24 48V DC power supplies

COMPLIANCE

Table 3. Compliance

Specification	Description
Regulatory compliance	Products should comply with CE Markings according to directives 2004/108/EC and 2006/95/EC.
Safety	UL 60950-1 Second Edition CAN/CSA-C22.2 No. 60950-1 Second Edition EN 60950-1 Second Edition IEC 60950-1 Second Edition AS/NZS 60950-1 GB4943
EMC: Emissions	47CFR Part 15 (CFR 47) Class A AS/NZS CISPR22 Class A CISPR22 Class A EN55022 Class A ICES003 Class A VCCI Class A EN61000-3-2 EN61000-3-3

	KN32 Class A CNS13438 Class A
EMC: Immunity	EN55024 CISPR24 EN300386 KN 61000-4 Series
RoHS	The product is RoHS-6 compliant with exceptions for leaded-ball grid-array (BGA) balls and lead press-fit connectors.

ORDERING INFORMATION

Table 4. Ordering information

Product number	Product description
8201-SYS	Cisco 8201 1 RU Chassis with 24x400 GbE QSFP56-DD and 12x100G QSFP28 and 32 GB DRAM
8202-SYS	Cisco 8202 2 RU Chassis with 12x400 GbE QSFP56-DD and 60x100 GbE QSFP28 and 32 GB of DRAM
8808-SYS	Cisco 8808 8-slot System
8812-SYS	Cisco 8812 12-slot System
8818-SYS	Cisco 8818 18-slot System

WHERE TO BUY

Want to buy this series of products? please contact:

- Tel: +1-626-239-8066 (USA) +852-3050-1066 / +852-3174-6166
- Fax: +852-3050-1066 (Hong Kong)
- Email: cisco@router-switch.com (Sales Inquiries)

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SOURCES

https://www.cisco.com/c/en/us/products/collateral/routers/8000-series-routers/datasheet-c78-742571.html